

Global Hackathon on Using AI for Digital Trade Regulatory Analysis

AI Tool Design

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UN ESCAP

Orientation Workshop | 1 June 2026 | 19:00- 21:00

A

The human data collection process

How data is collected today

B

Automation Workflow

Workflow + What the output must look like

C

Issues encountered with Automation

Based on experiences so far

D

Tool Design canvas

Walkthrough with a worked example

E

Take home briefing

Your assignment for next session

A · The human data collection process



Read and understand RDTII framework

RDTII 2.1 Guide
Understand Scope,
Coverage and unsaid
rules – legal
knowledge is implicit



Search

Search official
databases,
government websites,
and the web to find
primary legal sources.



Shortlist

Shortlist documents to
examine further
(possibly download).



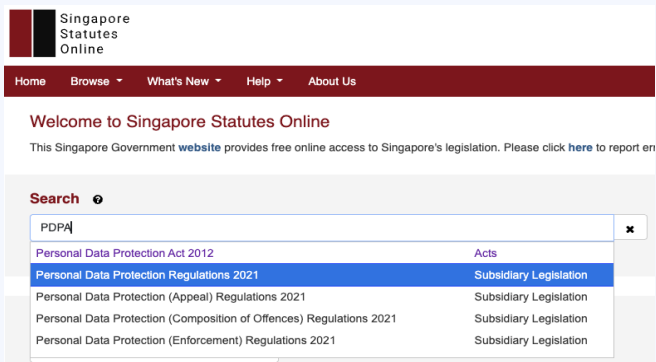
Retrieve & Record it

Read documents to
identify relevant
sections for RDTII
indicator (translation
required sometimes).



Score and Data verification

Manual verification
and then score as per
RDTII scoring criteria



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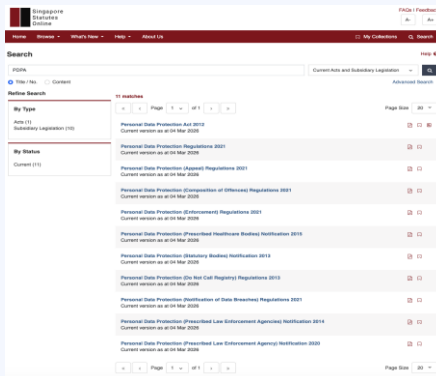
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Search

PDPA

Personal Data Protection Act 2012	Acts
Personal Data Protection Regulations 2021	Subsidiary Legislation
Personal Data Protection (Appeal) Regulations 2021	Subsidiary Legislation
Personal Data Protection (Composition of Offences) Regulations 2021	Subsidiary Legislation
Personal Data Protection (Enforcement) Regulations 2021	Subsidiary Legislation



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Personal Data Protection (Enforcement) Regulations 2021	Subsidiary Legislation

Transfer of personal data outside Singapore

26.—(1) An organisation must not transfer any personal data to a country or territory outside Singapore except in accordance with requirements prescribed under this Act to ensure that organisations provide a standard of protection to personal data so transferred that is comparable to the protection under this Act.

(2) The Commission may, on the application of any organisation, by written notice exempt the organisation from any requirement prescribed pursuant to subsection (1) in respect of any transfer of personal data by that organisation.

(3) An exemption under subsection (2) —

- (a) may be granted subject to such conditions as the Commission may specify in writing; and
- (b) need not be published in the *Gazette* and may be revoked at any time by the Commission.

(4) The Commission may at any time add to, vary or revoke any condition imposed under this section.

A · The human data collection process

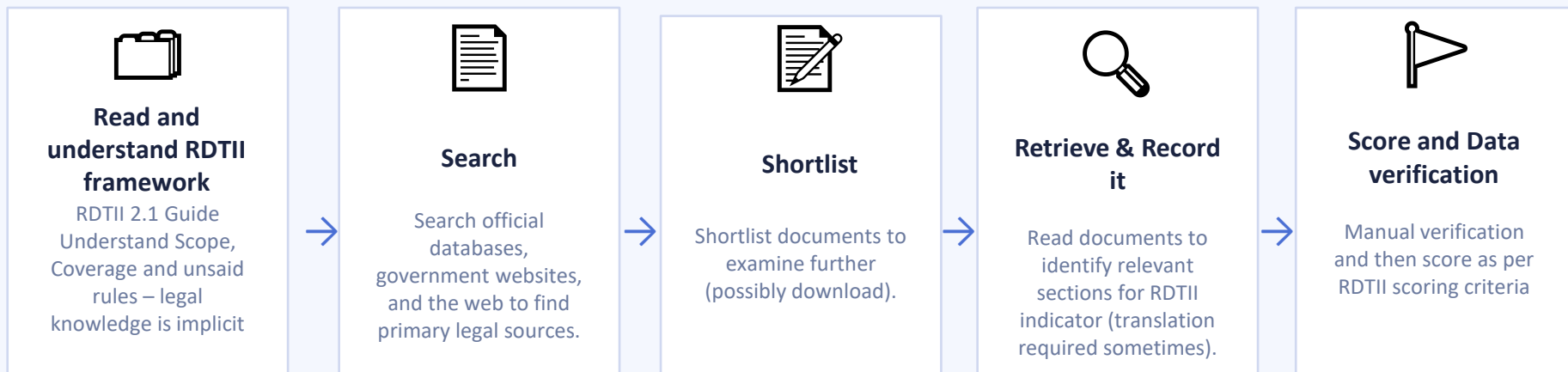
What does the database look like now? [This database is available on the knowledge portal]

RDTII pillar and indicators number (the methodology sheet in the excel has the full name and description of the indicators)

Pillar name and description of the indicators)			RDTII score	Name of the relevant Act/Law/Regulation	Sectoral/Horizontal	Explanation of how the recorded Act and/or practice relate to the RDTII indicator	Timeframe of the recorded Act and/or practice	URL to recorded Act and/or practice
Pillar_ID	Indicator_ID	Raw Score	Act and/or practice	Coverage	Impact or comments on Acts or practices	Timeframe	References	
6	6.1	0	Personal Data Protection Act 2012; Personal Data Protection (Amendment) Act 2020	Horizontal	Singapore does not implement ban on data transfer and local processing requirement.	Since 2012 last amended on 1 February 2021; Since February 2021	https://sso.agc.gov.sg/Act/PDPA2012 ; https://www.pdpc.gov.sg/Overview-of-PDPA/The-Legislation/Personal-Data-Protection-Act ; https://www.pdpc.gov.sg/overview-of-pdpc/the-legislation/personal-data-protection-act/data-protection-obligations	
6	6.2	0.5	Companies Act 1967	Accounting records	According to Section 199, every company must retain accounting for a period of 5 years from the end of the financial year in which the transactions or operations to which those records relate are completed. This data shall be kept at the registered office of the company or other places in Singapore specified by the director. This provision is to ensure that the company will have sufficient data to enable them to properly audited and true financial statements. Specifically, accounting records under this Act refers to records in relation to a corporation, includes such working papers and other documents as are necessary to explain the methods and calculations by which accounts of the corporation are made up (Section 4).	Since July 2014 last amended in 2021	https://sso.agc.gov.sg/Act/CoA1967	
6	6.3	0	Personal Data Protection Act 2012; Personal Data Protection (Amendment) Act 2020	Horizontal	Singapore does not implement infrastructure requirement.	Since 2012 last amended on 1 February 2021; Since February 2021	https://sso.agc.gov.sg/Act/PDPA2012	

This database will be a key resource for training your tool and cross-checking its results.

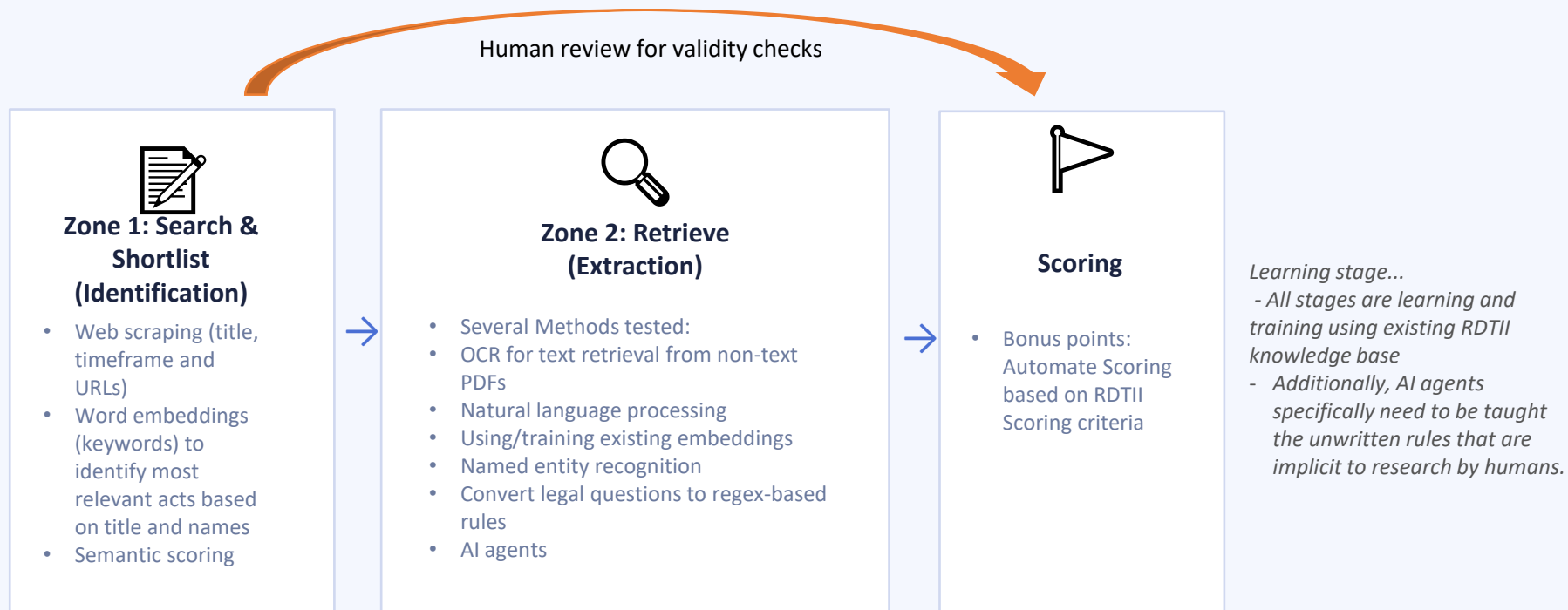
A · The human data collection process



Data collection is a manual and intensive process:

- More than 10 researchers per update
- Each country takes 1-4 weeks to do initial update, depending on complexity
- New countries surveyed + existing updated
- More than 2600 regulations across the region have been identified by researchers.

B · End to End Automation Workflow



Input

RDTII knowledge base

All resources available in

Knowledge portal

1. RDTII 2.1_ Round 1 Database.xlsx:
 1. Existing database
 2. URLs of official sources (if the country is already been surveyed before)
2. ESCAP-2025-RDTII-2.1- internal guide.pdf: Implicit rules
3. ESCAP-2025-RDTII-2.1-guide.pdf: complete methodological review of RDTII framework

Processing (AI Tool)

Zone 1: Identify legal/regulatory documents



Zone 2: Extract Legal text:

Example: Transfer of personal data outside Singapore (PDPA, 26.—(1) An organisation must not transfer any personal data to a country or territory outside Singapore except in accordance with requirements prescribed under this Act to ensure that organisations provide a standard of protection to personal data so transferred that is comparable to the protection under this Act.

(2) The Commission may, on the application of any organisation, by written notice exempt the organisation from any requirement prescribed pursuant to subsection (1) in respect of any transfer of personal data by that organisation.

(3) An exemption under subsection (2) —

(a) may be granted subject to such conditions as the Commission may specify in writing; and

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(4) The Commission may at any time add to, vary or revoke any condition imposed under this section.

Required Fields in Output (UI design can vary)

Provision text:

“An organisation must not transfer any personal data to a country or territory outside Singapore except in accordance with requirements prescribed under this Act”

Act/Practice(s)/Law Number:

Personal data protection Act (2012);
Personal data protection Regulation (2021)

Pillar:

Pillar 6 - Cross-border Data Policies

Indicator:

Indicator 6.4 – Conditional flow regime

Rational (optional):

The provisions prohibits transfer of personal data outside Singapore.

Coverage:

Horizontal/ Sectoral (Name of Sector):
In this case, Horizontal

Timeframe & Last amendment:

Since 2012; Since 2021

Address:

Article 26 (1)

URL:

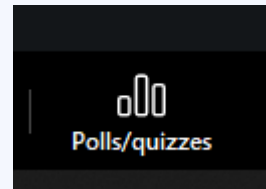
<https://sso.agc.gov.sg/Act/PDPA2012>
<https://sso.agc.gov.sg/SL/PDPA2012-563-2021>

Confidence/Relevance/Disclosure tag:

0-100% / 0 -1 / Low- High? Known-Unknown

Flag for review:

Yes/No



See polls/quizzes
section in Zoom!

In your view, what are the main challenge(s) in automating RDTII data collection?

A: Incorrect Legal Citations

B: Missing Act/Regulations

C: Misinterpretation of legal text

D: Incorrect response to core legal question of the indicator

E: All of the above



LET'S
VOTE

In your opinion, what are the main challenge(s) in automating RDTII data collection?

A: Incorrect Legal Citations

B: Missing Act/Regulations

C: Misinterpretation of legal text

D: Incorrect response to core legal question of the indicator

E: All of the above

ANSWER IS: ALL OF THE ABOVE AND THEN SOME MORE..

Issues automation tools encounter:

- Incorrect Legal Citation
- Missing Act/Regulation
- Misinterpretation of Law
- Incorrect response to core legal question of the indicator
- Broken/incorrect URLs
- Old documents pulled by tool; misses latest legal text
- Non-standard government portals, anti-bot barriers, JS rendering
- Unstructured formats: scanned PDFs, images, inconsistent layouts
- Hallucinations: Creating indicators not present in the RDTII framework or creating laws that don't exist

Singapore. Pillar 6 - Indicator 6.1: Ban and local processing requirements

URL: <https://sso.agc.gov.sg/Act/BA> **Title:** Banking Act (Cap. 19) (Section 47) **Relevance Score:** 0.70

Text: > Except as expressly provided in the Act, no person who is or has been an officer of a bank shall disclose any customer information to any other person. Bank customer information is to be **kept confidential** and must not be divulged to any unauthorized party.

Relevance Notes

This banking law provision effectively prohibits the transfer or disclosure of bank customer data to any external party (including overseas recipients) except under authorized circumstances. **It serves as a strict confidentiality/local handling requirement for financial customer data.**

Why is this an incorrect output from automation tool? Choose your answer in the poll!

- A. Misinterpretation of Law
- B. Broken/incorrect URLs
- C. Incorrect Legal Citation

Issues automation tools encounter:

- Incorrect Legal Citation
- Missing Act/Regulation
- Misinterpretation of Law
- Incorrect response to core legal question of the indicator
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- Old documents pulled by tool; misses latest legal text
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Relevance Notes

This banking law provision effectively prohibits the transfer or disclosure of bank customer data to any external party (including overseas recipients) except under authorized circumstances. It serves as a strict confidentiality/local handling requirement for financial customer data.

Misinterpretation of law: 1. The Banking Act is a binding domestic law, but the cited provision does not impose a ban on cross-border data transfers or a local processing requirement. It establishes a general duty of confidentiality applicable to all disclosures, domestic or cross-border, subject to statutory exceptions.

2. mischaracterisation of a confidentiality rule as a localisation or transfer ban.

D • Tool Design Canvas

- 1 **Zone**
- Which RDTII automation zone this tool/stage targets. Zone 1: Search and shortlist or Zone 2: Retrieval or BOTH

- 3 **Outputs ★**
- Exact structure the tool produces (certain fields are compulsory but final UI and design is upto you!)

- 5 **Risks**
- Hallucination, bias, missed provisions (as per list in slide 10 and more)

- 2 **Inputs ★**
- Data going in: source, format, language

- 4 **Evaluation ★**
- Metrics, test set, human validation

7

8

D · Worked example 1 — Zone 1 · Identification

Zone	Zone 1 Identification																																																
Inputs ★	Malaysia + description of pillar 2: Public Procurement																																																
Outputs ★	<table><tr><th>Title</th><th>Status</th><th>final_rank</th><th>ranking_score</th><th>semantic</th><th>keyword</th><th>keyword</th><th>PDF_URL</th></tr><tr><td>FEDERAL TERRITORY (FINANCIAL ARRANGEMENT) ACT 1982</td><td>Acts</td><td>0.70831</td><td>0.7083</td><td>0.9583</td><td>0.3333</td><td>1</td><td>https://loi</td></tr><tr><td>STATE WATER SUPPLY FUND (FINANCIAL AND ACCOUNTING PROCEDURE) ACT 1980</td><td>Acts</td><td>0.70757</td><td>0.7076</td><td>0.9571</td><td>0.3333</td><td>1</td><td>https://loi</td></tr><tr><td>FINANCIAL PROCEDURE ACT 1957</td><td>Acts</td><td>0.707318</td><td>0.7073</td><td>0.9566</td><td>0.3333</td><td>1</td><td>https://loi</td></tr><tr><td>FINANCIAL REPORTING (AMENDMENT) ACT 2012</td><td>Acts</td><td>0.707056</td><td>0.7071</td><td>0.9562</td><td>0.3333</td><td>1</td><td>https://loi</td></tr><tr><td>NETTING OF FINANCIAL AGREEMENTS ACT 2015</td><td>Acts</td><td>0.70703</td><td>0.707</td><td>0.9562</td><td>0.3333</td><td>1</td><td>https://loi</td></tr></table> *This is a real result from our trials: Web scraping Malaysia law repository for Pillar 2 Using a combination of Semantic Analysis + keyword matching	Title	Status	final_rank	ranking_score	semantic	keyword	keyword	PDF_URL	FEDERAL TERRITORY (FINANCIAL ARRANGEMENT) ACT 1982	Acts	0.70831	0.7083	0.9583	0.3333	1	https://loi	STATE WATER SUPPLY FUND (FINANCIAL AND ACCOUNTING PROCEDURE) ACT 1980	Acts	0.70757	0.7076	0.9571	0.3333	1	https://loi	FINANCIAL PROCEDURE ACT 1957	Acts	0.707318	0.7073	0.9566	0.3333	1	https://loi	FINANCIAL REPORTING (AMENDMENT) ACT 2012	Acts	0.707056	0.7071	0.9562	0.3333	1	https://loi	NETTING OF FINANCIAL AGREEMENTS ACT 2015	Acts	0.70703	0.707	0.9562	0.3333	1	https://loi
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Evaluation ★	Human spot-check, Inter-Annotator Agreement																																																
Risks	Because 1-2 key word match, the tool will push an irrelevant act to the top – Water supply fund is NOT relevant at all and it has a high score because the act has the words financial and accounting in its name																																																

D · Worked example 2 — Zone 1 · Identification

Zone	Zone 1 Identification
Inputs ★	Singapore + description of pillar 4: Intellectual Property Rights (Indicator 4.2: Patent enforcement issues: civil and administrative procedures and remedies; and provisional measures)
Outputs ★	1. Patents Act 1994 — Section 69 (Restrictions on Relief for Infringement) a) Instrument Citation • Title: Patents Act 1994 (2021 Revised Edition) • Type: Act (Primary Legislation) • Issuing Authority: Parliament of Singapore • Section / Article / Clause: Part 13, Section 69 — “Restrictions on relief for infringement” • Effective Date: 26 February 1995 (original commencement); current consolidated version as of 31 December 2021 (2021 Revised Edition) • In-force status: Confirmed in force as of 2026; current version maintained at sso.agc.gov.sg (last updated 6 March 2026) • Official URL: https://sso.agc.gov.sg/Act/PA1994 *This is a real result from our trials: Using LLM to extract relevant law from Singapore Law database
Evaluation ★	Human spot-check, Inter-Annotator Agreement
Risks	The effective date field can have inaccuracies due to difficulties crawling Singapore law website, JS rendering and anti bot measures. In this case, revised edition is wrong: It should be Amended by Act 5 of 2025 on 9th March 2025 .

D · Worked example 3 — Zone 2 · Extraction

Zone	Zone 2 Extraction
Inputs ★	Thailand PDPA B.E. 2562, Section 28 : In the event that the Data Controller sends or transfers the Personal Data to a foreign country, the destination country or international organization that receives such Personal Data shall have adequate data protection standard, and shall be carried out in accordance with the rules for the protection of Personal Data as prescribed by the Committee in section 16 (5), except in the following circumstances: + RDTII knowledge base
Outputs ★ (ignore UI and design of how output will be presented for now)	<pre>JSON record: { "Pillar_tag": Pillar 6 "indicator_tag": "Indicator 6.4: Conditional Flow Regimes" "obligation": "no cross-border transfer without have adequate data protection standard ", "binding": true, "exceptions": ["contract performance", "explicit consent"], "key_text": "shall not send or transfer...", "source": "Thailand PDPA B.E. 2562, Section.28" , "URL": "https://pdpathailand.com/pdpa/index_eng.html?srsltid=AfmBOoonwgXYKkleZQFiInIU4rybk2fw53n9chHj1nz2uOiVaZc693g3" "Coverage": "Horizontal" "Time frame": "Since 2019" }</pre>
Evaluation ★	Human spot-check, field accuracy
Risks	Misinterpretation of law, Thai year might not be correctly interpreted as 2019, coverage might be wrong, JSON record not understandable to non-technical person, user interface necessary



Take home assignments!

Submit your assignments before 4th June
2026, 16:00 Bangkok Time

Email: escap-digitaltrade-hackathon@un.org

We will provide feedback.
Assignments available on Knowledge portal:

Scan to request access!



Your take home assignment 1

Why is this an incorrect output from automation tool?

Submit your answer before 4th June 2026, 16:00 Bangkok Time Email: escap-digitaltrade-hackathon@un.org

A reminder: Issues automation tools encounter:

- Incorrect Legal Citation
- Missing Act/Regulation
- Misinterpretation of Law
- Incorrect response to core legal question of the indicator
- Broken/incorrect URLs
- Outdated laws/regulations pulled by tool
- Non-standard government portals, anti-bot barriers, JS rendering
- Unstructured formats: scanned PDFs, images, inconsistent layouts
- Hallucinations: Creating indicators not present in the RDTII framework

Singapore. Pillar 7 - Indicator 7.2: Lack of dedicated legal framework for cybersecurity

URL: <https://www.mas.gov.sg/regulation/notices-and-guidelines/cyber-hygiene-notices-2019> **Title:** MAS Notice on Cyber Hygiene (2019) **Type:** other **Relevance Score:** 0.85

Text: MAS introduced legally binding Notices on Cyber Hygiene in 2019, requiring all financial institutions to implement a set of baseline cybersecurity measures. For example, FIs must establish robust access controls (including multi-factor authentication), promptly apply security patches to systems, deploy malware protection, and maintain network security monitoring. These requirements are mandatory and enforceable across the financial sector.

Relevance Notes

This evidence shows a dedicated cybersecurity framework at the sectoral level. The financial sector regulator (MAS) has imposed specific cybersecurity obligations on financial institutions, demonstrating that beyond the general cybersecurity law, Singapore enforces cybersecurity requirements through sector-specific regulations.

Your take home assignment 2

- 1 Choose one example from Zone 1 and one from Zone 2, and provide details for upto two alternative tool design concepts. I have provided one example each in the template sheet – you are free to make your own.
- 2 Fill in all 5 sections of the Tool Design Canvas (★ sections are non-negotiable)
- 3 Manually walk your example through your own described process: write out what the output would look like
- 4 Identify the 1-2 risks you are most worried about

Submit word doc or PDF of your completed canvas before 4th June 2026, 16:00 Bangkok Time. I will provide feedback during workshop 2 on 5th June 2026.

Email: escap-digitaltrade-hackathon@un.org

Your take home assignment 2

ESCAP x [EMITL](#) | AI Hackathon for Digital Trade Regulation

Tool Design Canvas Workshop 1 June 2026 19:00 BKK time

Submit completed template before 4th June 16:00 BKK time

Zone	Input*	Output*	Evaluation*	Key Risk
Which automation tool? Zone 1 (preferred) Zone 2 (extraction) or both	What does the tool receive? Indicator + economy name or raw provision text or both	What does the tool produce? Ranked list, JSON record, annotated text, etc. Specify exact format	How will you know it works? F1, field accuracy, inter-annotator agreement or others	What is the biggest risk? Hallucination, relevance drift, scope-sweep, data quality, legal ambiguity
Tool concept option 1 Example - Zone 1: Retrieve a shortlist of regulations for Malaysia (potentially) relevant to Pillar 7: Domestic Data protection & Privacy				
Tool concept option 2				
Tool concept option 3 Example - Zone 2: Extract relevant provisions and details for Pillar 7 indicator(s) for Malaysia from within legal text.				
Tool concept option 2				

This canvas is a planning tool — not a submission form. One row per tool concept your team considers.

Submit word doc or PDF of your completed canvas before 4th June 2026, 16:00 Bangkok Time. I will provide feedback during workshop 2 on 5th June 2026.

Email: escap-digitaltrade-hackathon@un.org



ESCAP
Economic and Social Commission
for Asia and the Pacific

Global Hackathon on Using AI for Digital Trade Regulatory Analysis

**Thank you! Looking
forward to seeing your
tool design canvas.**

For any queries reach me at nikita.shahu@un.org

Orientation Workshop | 1 June 2026 | 19:00- 21:00